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EXPERIENCE

HashedIn by Deloitte

Software Engineer 1

On-Site - Chennai, India

Nov 2025 – Present

Autonomous Coding

- Architected an **Agentic AI** system using **Django**, reducing manual generation time for XML, XSLT, SQL, and Oracle Fast Formulae by **~70%** across internal development workflows.
- Orchestrated multi-agent logic to translate natural language and mathematical functions into Oracle Fast Formulae, owning end-to-end accuracy for **10+** formula types via **Azure Cosmos DB**-backed DBI retrieval.
- Integrated **Azure OpenAI** and **Anthropic** models to enforce rule-based flow control and contextual data validation, directly reducing formula error rates by **~40%** across the pipeline.
- Engineered full-cycle generation of Oracle eText and RTF templates, taking ownership of automated conditional formatting and logic validation across template types to ensure production-ready output.
- Implemented **RAG-based retrieval** to power company-specific Q&A and contextual validation, cutting domain-sensitive query resolution time by **~60%** within the automation pipeline.

Multi-Agent Code Analysis System

- Architected a **Multi-Agent** System using **LangGraph** to automate codebase analysis, generating role-specific documentation for Software Engineers (technical) and Product Managers (business-logic).
- Developed an Interactive Agentic Pipeline with **FastAPI**, enabling real-time state management and custom instruction injection, reducing analysis re-runs by **~50%** through mid-pipeline steering.
- Engineered Semantic Code Retrieval using **pgvector** within a **PostgreSQL** (pgAdmin) environment to chunk, store, and query code metadata to deliver high-context agent responses with **~80%** retrieval accuracy.
- Integrated Agents using **LangChain** and **RAG** principles to search for external documentation and best practices, ensuring the system provides up-to-date analysis of frameworks and libraries.
- Built a Real-Time Monitoring Dashboard in **Streamlit** that visualizes agent progress, provides a project-based user interface, and enables natural language Q&A over the analysed codebase.

Microchip Technology

Engineer I - Corporate Applications Engineer

On-Site - Chennai, India

Jul 2024 – Mar 2025

- Worked with clients to deliver critical project tasks across the entire WNET (L1 and L2) product portfolio, consistently meeting deadlines.
- Developed a **Python-based tool using OpenCV** to extract pin names from customer schematics, significantly accelerating design validation and enabling usage across global engineering teams.
- Designed and implemented a **custom network testing tool in Python**, tailored for proprietary hardware platforms. The tool improved test efficiency and reduced manual effort by **30%**.
- Designed and executed **functional and protocol tests** for wired network products with integrated MPUs on Linux, using custom **Python scripts and libraries** to accelerate validation workflows.
- Developed a **multi-screen GUI application (6 screens)** with structured modular design, and packaged it into a standalone executable. Conducted comprehensive testing to ensure seamless functionality and deployment.
- Validated and analyzed the implementation of IEEE protocols such as PTP and ACL on managed Ethernet switches.

Student Intern

Microchip Technology

Chennai, India

Jan 2024 - Jun 2024

- Designed and implemented an embedded C solution to enable Ethernet communication on entry-level microcontrollers lacking native Ethernet support, achieving a **25% improvement in data transmission speed** through software optimization.
- Developed a **lightweight software utility** for an **hourly reminder system** with configurable intervals, optimized for low-power operation on resource-constrained embedded platforms, ensuring efficient performance and extended runtime.

Workshop

Bangalore, India (Sep 2022)

- Gained hands-on experience with Android development tools and implemented real-time text-to-speech functionality conducted by **IISc Bangalore** to enhance practical knowledge in mobile app development.

EDUCATION

Thiagarajar College of Engineering Bachelors of Engineering in Electronics and Communication Engineering	Madurai, India Sep 2020 – May 2024 CGPA 9.61
Vidhya Sagar Global school 12 th Grade	Chengalpattu, India June 2019 - March 2020 Score 92%

PROJECTS

- **WhisperQL:** Built a web application featuring **FastAPI** backend and **Streamlit** frontend that allows users to connect to a **PostgreSQL** database and generate executable SQL queries from **plain-English questions** through an intuitive dark-themed interface. Integrated multiple **LLM providers (Gemini, Claude, GPT)** with automatic schema introspection, with a pre-execution review gate for safety.
- **Email Automation Script:** Developed a **Python script using Selenium WebDriver** to automate email sending for a college club event, successfully delivering over **500 invitation emails** to participants.
- **Car Showroom Management System:** Developed a desktop-based application with full **CRUD** functionality to manage vehicle inventory, track status updates, and streamline overall showroom operations.
- **Hostel Management System:** Built a web-based student record management system with complete **CRUD** operations, integrating a relational database for efficient data storage, retrieval, and management.
- **Smog Prediction and alerting system:** Executed end-to-end integration of a smog detection system using NodeMCU, environmental sensors, and custom PCB for interfacing. Enabled real-time and cloud-based monitoring with ThingSpeak, applying data analytics and **predictive modeling (LPC)** for trend analysis. Combined hardware design, real-time data processing, and IoT for smart environmental monitoring.
- **Hobby Projects:**
 - Designed and implemented game logic for Tic-Tac-Toe and Black Jack using **Python**.
 - Implemented web scraping projects using **Python**, leveraging libraries like Requests, BeautifulSoup, and CSV for data extraction and organization.

SKILLS

- **Languages:** Python, Java, JavaScript, PHP, SQL, C
- **Frameworks:** FastAPI, Django, Streamlit
- **Database:** PostgreSQL, Azure CosmosDB
- **Generative AI:** Langchain, Langgraph, Autogen, OpenAI, Anthropic
- **Version Control:** Git
- **Web Technologies:** HTML, CSS

CERTIFICATIONS

- Azure AI Fundamentals - Microsoft
- Django – The complete course (Beginner + Advanced + AI) - Udemy
- Building AI Agents & Agentic AI system via Microsoft Autogen – Udemy
- Claude with Amazon Bedrock - Anthropic
- FastAPI – The complete course (Beginner + Advanced) – Udemy
- Taming Big Data with Apache Spark and Python - Udemy
- The complete 2024 Web Development Bootcamp - Udemy.